

Mixture & Ratio – Solutions & Answers

Detailed step-by-step solutions for all 20 questions.

Question 1

A certain amount of water was poured into a 300 litre container and the remaining portion of the container was filled with milk. Then an amount of this solution was taken out which was twice the volume of water initially poured, and water was poured to refill. If the resulting solution contains 72% milk, the amount of water initially poured is:

Correct Answer: 30

Solution:

- Let initial water = W litres. Initial milk $M = 300 - W$.
 - Volume of solution removed = $2W$. Water is added back ($2W$ litres).
 - Final milk = $(300 - W) \times (1 - 2W/300) = (300 - W)(300 - 2W)/300$.
 - 72% milk means final milk = $0.72 \times 300 = 216$ litres.
 - $(300 - W)(300 - 2W) = 216 \times 300 = 64800$.
 - Expanding: $90000 - 600W - 300W + 2W^2 = 64800$.
 - $2W^2 - 900W + 25200 = 0 \rightarrow W^2 - 450W + 12600 = 0$.
 - Testing $W = 30$: $900 - 13500 + 12600 = 0$. ✓
 - The other root is 420, which exceeds 300 (container volume), so it is rejected.
 - Answer: $W = 30$ litres.
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Question 2

Anil mixes cocoa with sugar 3:2 (mixture A), coffee with sugar 7:3 (mixture B). A and B combined 2:3 to form C. C mixed with equal milk. % sugar in drink?

Correct Answer: A) 17%

Solution:

- Sugar in A = $2/(3+2) = 2/5$.
 - Sugar in B = $3/(7+3) = 3/10$.
 - Sugar in C (A:B = 2:3): $(2 \times 2/5 + 3 \times 3/10)/(2+3) = (4/5 + 9/10)/5 = (8/10 + 9/10)/5 = 17/50$.
 - Mixing C with equal milk (0% sugar) in 1:1 ratio:
 - Sugar in drink = $(1 \times 17/50 + 1 \times 0)/(1+1) = 17/100 = 17\%$.
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Question 3

Amal:Bimal scores initially 11:14. After equal increase, ratio becomes 47:56. Find ratio of Bimal's new score to his original.

Correct Answer: A) 4 : 3

Solution:

- Let original scores be $11k$ and $14k$; let increase = x .
 - $(11k + x)/(14k + x) = 47/56$.
 - Cross-multiplying: $56(11k + x) = 47(14k + x)$.
 - $616k + 56x = 658k + 47x \rightarrow 9x = 42k \rightarrow x = 14k/3$.
 - Bimal's new score = $14k + 14k/3 = 56k/3$.
 - Ratio = $(56k/3) / 14k = 56/42 = 4/3$.
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- Answer: 4 : 3.
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Question 4

20% ethanol mixed with solution S (concentration s%) in ratio 1:3. This mixture then mixed with equal volume of 20% ethanol → 31.25%. Find s.

Correct Answer: C) 50%

Solution:

- Step 1: Mix 1L of 20% with 3L of s%. Total = 4L.
 - Ethanol = $0.20 + 3s/100$. Concentration M■ = $(0.20 + 0.03s)/4 \times 100 = 5 + 0.75s$ %.
 - Step 2: Mix 4L of M■ with 4L of 20%. Total = 8L.
 - Total ethanol = $(5+0.75s)/100 \times 4 + 0.20 \times 4 = (0.20 + 0.03s) + 0.80 = 1.00 + 0.03s$.
 - Final concentration = $(1.00 + 0.03s)/8 \times 100 = 31.25$.
 - $1.00 + 0.03s = 2.5 \rightarrow 0.03s = 1.5 \rightarrow s = 50\%$.
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Question 5

Jar: 175 ml water + 700 ml alcohol = 875 ml. Gopal removes 10% and replaces with water — twice. Final % of water?

Correct Answer: B) 35.2%

Solution:

- Initial alcohol = 700, water = 175. Total = 875. Initial alcohol % = $700/875 = 80\%$.
 - Each operation: 10% removed and replaced with water. Fraction retained = 0.9.
 - Alcohol % after 2 operations = $80\% \times 0.9 \times 0.9 = 80\% \times 0.81 = 64.8\%$.
 - Final water % = $100\% - 64.8\% = 35.2\%$.
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Question 6

Drum 1: A:B = 18:7. Drums mixed 3:4. Final A:B = 13:7. Find ratio A:B in drum 2.

Correct Answer: B) 239 : 161

Solution:

- Concentration of A in drum 1: $C■ = 18/25$.
 - Let concentration of A in drum 2 = $C■$.
 - Final concentration = $13/(13+7) = 13/20$.
 - Weighted average: $(3 \times 18/25 + 4 \times C■)/(3+4) = 13/20$.
 - $(54/25 + 4C■)/7 = 13/20 \rightarrow 54/25 + 4C■ = 91/20$.
 - $4C■ = 91/20 - 54/25 = 455/100 - 216/100 = 239/100$.
 - $C■ = 239/400$, so $x = 239$ and $x+y = 400$, giving $y = 161$.
 - Ratio A:B in drum 2 = 239 : 161.
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Question 7

Two 800 cc bottles: 33% and 17% indigo. Some solution from bottle 1 replaced with equal volume from bottle 2 → bottle 1 now 21%. Volume left in bottle 2?

Correct Answer: 200 cc

Solution:

- Using alligation: ratio of 33% solution to 17% solution in final mix = $(21-17):(33-21) = 4:12 = 1:3$.

- Total 800 cc in bottle 1: $V_{\text{milk}} (33\%) = \frac{1}{4} \times 800 = 200 \text{ cc}$; $V_{\text{water}} (17\% \text{ added}) = \frac{3}{4} \times 800 = 600 \text{ cc}$.
 - So 600 cc was transferred from bottle 2 to bottle 1.
 - Volume left in bottle 2 = $800 - 600 = 200 \text{ cc}$.
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Question 8

Glass: 500 cc milk. Cup: 500 cc water. Transfer 150 cc milk to cup, mix. Transfer 150 cc back to glass. Ratio of water in glass to milk in cup?

Correct Answer: A) 1:01 (1:1)

Solution:

- After step 1: Glass has 350M, 0W. Cup has 150M + 500W = 650 total. Milk conc in cup = $150/650 = 3/13$.
 - Step 2: Transfer 150 cc from cup to glass.
 - Water in 150 cc = $(500/650) \times 150 = 1500/13 \text{ cc}$.
 - Milk in 150 cc = $(150/650) \times 150 = 450/13 \text{ cc}$.
 - Water in glass = $1500/13 \text{ cc}$.
 - Milk left in cup = $150 - 450/13 = (1950-450)/13 = 1500/13 \text{ cc}$.
 - Ratio = $(1500/13) : (1500/13) = 1:1$.
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Question 9

Container: 40 L milk. Remove 4 L, replace with water repeatedly. Smallest n where milk volume < water volume?

Correct Answer: 7

Solution:

- After n operations, milk = $40 \times (9/10)^n$.
 - Condition: milk < water $\rightarrow$ milk < 20 $\rightarrow 40 \times (0.9)^n < 20 \rightarrow (0.9)^n < 0.5$.
 - n=1: 0.9; n=2: 0.81; n=3: 0.729; n=4: 0.6561; n=5: 0.59; n=6: 0.53; n=7: 0.478 < 0.5. ✓
 - Answer: 7.
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Question 10

40 L solution, dye:water = 2:3. Water added $\rightarrow$ 2:5. Then 1/4 removed. How many litres of dye needed to restore 2:3?

Correct Answer: 8

Solution:

- Initial: dye = 16 L, water = 24 L. Total = 40 L.
 - After water added to make 2:5: dye stays 16 L, water = $16 \times 5/2 = 40 \text{ L}$. Total = 56 L.
 - Remove 1/4 of 56 L = 14 L. Remaining = 42 L.
 - Remaining dye = $(2/7) \times 42 = 12 \text{ L}$; water = $(5/7) \times 42 = 30 \text{ L}$.
 - To restore 2:3: $(12+d)/30 = 2/3 \rightarrow 12+d = 20 \rightarrow d = 8 \text{ L}$.
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Question 11

Alloy: A:B:C = 3:4:7 by volume. Weight per same volume A:B:C = 5:2:6. In 130 kg alloy, weight of C?

Correct Answer: B) 84 kg

Solution:

- Weight ratio = volume $\times$ density.

- A: $3 \times 5 = 15$; B: $4 \times 2 = 8$; C: $7 \times 6 = 42$. Weight ratio = 15:8:42. Sum = 65.
 - Weight of C = $(42/65) \times 130 = 42 \times 2 = 84$ kg.
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Question 12

Rajesh: 20 ha, Vimal: 30 ha. Vimal wheat:mustard = 5:3. Total wheat:mustard = 11:9. Rajesh's wheat:mustard?

Correct Answer: B) 7:9

Solution:

- Vimal: $W_V = 5/8 \times 30 = 18.75$ ha; $M_V = 3/8 \times 30 = 11.25$ ha.
 - Total 50 ha: $W_{Total} = 11/20 \times 50 = 27.5$ ha; $M_{Total} = 9/20 \times 50 = 22.5$ ha.
 - Rajesh: $W_R = 27.5 - 18.75 = 8.75$ ha; $M_R = 22.5 - 11.25 = 11.25$ ha.
 - Ratio = $8.75 : 11.25 = 875 : 1125 = 35:45 = 7:9$.
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Question 13

Vessel: acid + water. Add 2 L water $\rightarrow$ 50% acid. Add 15 L acid $\rightarrow$ 80% acid. Original water:acid ratio?

Correct Answer: B) 3 : 5

Solution:

- Let acid = A, water = W initially.
 - After +2L water: $A/(A+W+2) = 0.5 \rightarrow 2A = A+W+2 \rightarrow A = W+2 \dots(1)$
 - After +15L acid: $(A+15)/(A+W+17) = 0.8 \rightarrow A+15 = 0.8A+0.8W+13.6$.
 - $0.2A - 0.8W = -1.4 \rightarrow A - 4W = -7 \dots(2)$
 - Substitute (1) into (2): $(W+2) - 4W = -7 \rightarrow -3W = -9 \rightarrow W = 3, A = 5$.
 - Original water:acid = 3:5.
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Question 14

When Rajesh's age = Garima's current age, Rajesh:Garima = 3:2. When Garima's age = Rajesh's current age, ratio will be?

Correct Answer: C) 5 : 4

Solution:

- Let Rajesh's current age = R, Garima's = G. Let $R = G + x$ (x = age difference).
 - x years ago, Rajesh was G, Garima was $G - x$. Ratio $G/(G-x) = 3/2$.
 - $2G = 3G - 3x \rightarrow G = 3x, R = 4x$.
 - In x years (when Garima = R): Rajesh = $5x$, Garima = $4x$.
 - Future ratio = $5x : 4x = 5 : 4$.
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Question 15

Two containers, each volume V, half-filled with sugar syrup (C1) and milk (C2). Three transfer operations. Final SS:milk in C2?

Correct Answer: D) 5:6

Solution:

- Let V = total container volume. C1 initially: $V/2$ SS. C2 initially: $V/2$ Milk.
- Step 1: Transfer $V/4$ (half of C1) to C2. C1: $V/4$ SS. C2: $V/4$ SS + $V/2$ M (total $3V/4$, SS:M = 1:2).
- Step 2: Transfer $3V/8$ (half of C2) back to C1. SS transferred = $V/8$, M = $V/4$. C2: $V/8$ SS, $V/4$ M.

- Step 3: Transfer $5V/16$ (half of $C1$) to $C2$. $C1$ after step 2 has $3V/8$ SS; $SS:M = 3:2$ (approx $3V/8$ total).
- After all transfers, final $C2$: $SS = 5V/16$, Milk = $6V/16$. Ratio = $5:6$.

Question 16

Milk container: draw 9 L twice, replace with water each time. Final milk:water = $16:9$. Capacity?

Correct Answer: 45 litres

Solution:

- Let capacity = V . After 2 replacements, milk = $V \times (1 - 9/V)^2$.
- Milk fraction = $16/(16+9) = 16/25$.
- $(1 - 9/V)^2 = 16/25 \rightarrow 1 - 9/V = 4/5 \rightarrow 9/V = 1/5 \rightarrow V = 45$.

Question 17

Initial alloy (weight W , silver fraction x) + 3 kg pure silver $\rightarrow$ 90% silver. Same alloy + 2 kg of 90% silver alloy $\rightarrow$ 84% silver. Find W .

Correct Answer: C) 3 kg

Solution:

- Scenario 1: $(Wx + 3)/(W + 3) = 0.9 \rightarrow Wx + 3 = 0.9W + 2.7 \rightarrow Wx - 0.9W = -0.3 \dots(1)$
- Scenario 2: Silver added = 1.8 kg. $(Wx + 1.8)/(W + 2) = 0.84$.
- $Wx + 1.8 = 0.84W + 1.68 \rightarrow Wx - 0.84W = -0.12 \dots(2)$
- Subtract (1) from (2): $0.06W = 0.18 \rightarrow W = 3$ kg.

Question 18

Village: males:females = $5:4$. Literate $M:F = 2:3$. Illiterate $M:F = 4:3$. Literate males = 3600. Total females?

Correct Answer: 43200

Solution:

- Let literate males = $2y = 3600 \rightarrow y = 1800$. Literate females = $3y = 5400$.
- Let illiterate males = $4z$, illiterate females = $3z$.
- Total males: $5x = 3600 + 4z \dots(1)$. Total females: $4x = 5400 + 3z \dots(2)$.
- Multiply (1) by 3 and (2) by 4: $15x = 10800 + 12z$; $16x = 21600 + 12z$.
- Subtract: $x = 10800$. Total females = $4x = 43200$.

Question 19

Amal buys 110 kg syrup + 120 kg juice (syrup 20% cheaper). Sells 10 kg syrup at 10% profit, 20 kg juice at 20% profit. Remaining mixture sold at ■308.32/kg, overall 64% profit. Cost of syrup per kg?

Correct Answer: ■160

Solution:

- Let juice CP = J /kg. Syrup CP = $0.8J$ /kg.
- Total CP = $110 \times 0.8J + 120 \times J = 88J + 120J = 208J$.
- Revenue from initial sales: $10 \times 0.8J \times 1.1 + 20 \times J \times 1.2 = 8.8J + 24J = 32.8J$.
- Remaining: 100 kg syrup + 100 kg juice = 200 kg. Revenue = $200 \times 308.32 = \blacksquare 61664$.
- Total SP = $32.8J + 61664$. Overall profit 64%: Total SP = $1.64 \times 208J = 341.12J$.
- $32.8J + 61664 = 341.12J \rightarrow 308.32J = 61664 \rightarrow J = 200$.
- Syrup CP = $0.8 \times 200 = \blacksquare 160$ /kg.

Question 20

Initial mixture: lemon juice : sugar syrup = 1:1. New mixture formed by combining original mixture and sugar syrup in ratio 1:3. Final ratio of lemon juice : sugar syrup?

Correct Answer: D) 1 : 7

Solution:

- Take 1 unit of initial mixture: Lemon = 0.5, Sugar = 0.5.
 - Add 3 units of pure sugar syrup.
 - Total Lemon = 0.5, Total Sugar = $0.5 + 3 = 3.5$.
 - Ratio = $0.5 : 3.5 = 1 : 7$.
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